



## IDENTIFICATION

CODE : GI-5-S1-EC-IFD  
ECTS : 3

## HOURS

Cours : 0h  
TD : 26h  
TP : 0h  
Projet : 0h  
Evaluation : 2h  
Face à face pédagogique : 28h  
Travail personnel : 30h  
Total : 58h

## ASSESSMENT METHOD

## TEACHING AIDS

James A. Tompkins, 2010, Facilities Planning, John Wiley & Sons Inc  
Mikell P. Groover, 2016, Work Systems: The Methods, Measurement & Management of Work, PEARSON edition

## TEACHING LANGUAGE

English

## CONTACT

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## AIMS

## CONTENT

- 1- Introduction to facility design
  - a. Relationship between the product, process and scheduling design and the facility design
  - b. Flow system (Flow shop, Job Shop, cellular manufacturing, Fixed-position)
  - c. Process modeling and material flow analysis (Process Flowchart, IDEF0)
  - d. Activity relationship
- 2- Requirements of Manufacturing Facility (plant) design
  - a. Motion and Time study
  - b. Capacity determination (machine, human resource)
  - c. Performance measurement
- 3- Material Handling system, Storage and Warehouse system design
  - a. Type of material handling (receiving and shipping, dock equipment, moving equipment)
  - b. Type of Storage and Warehousing (Picking cart, Picking truck, vertical Warehouse, Automated Warehouse)
  - c. Transfer Line (Straight Line Flow, Circular Flow, U-Flow, Spine-Flow)
  - d. Equipment selection and Capacity determination
- 4- Cellular Manufacturing System (CMS) design
  - a. Clustering approach (Rank ordering, clustering, similarity coefficient, mathematical programming)
  - b. Designing and planning issues of CMS (capacity, material handling, layout)
- 5- Facility Layout
  - a. Basic layout types
  - b. Multi-floor facility layout,
  - c. Service layout
  - d. Flow shop and line balancing
  - e. Job shop and department location problem
  - f. Development alternatives
  - g. Mathematical modeling and Algorithms
- 6- Flexible Manufacturing System (FMS) design and optimization: (types of flexibilities, flexibilities classification, flexibility modeling and measurement, decision-making and optimization of the level of flexibility)
- 7- Industry 4.0 (Cyber Physical system, IIoT, Data Analytics and prescription of decision)
- 8- Configuration and experimentation of a flexible manufacturing system: Practical work on the Smart Factory platform (cyber-physical system).

## BIBLIOGRAPHY

S. Heragu, 2008, Facilities Design, CRC Press Inc; Third Edition,  
Ralph M. Barnes, Motion and Time Study: Design and Measurement of Work, 1980  
Alberto Garcia-Diaz & J. MacGregor Smith, 2024, Facilities Planning and Design, Springer

## PRE-REQUISITES

Mathematical modeling and optimization, Statistic and Probability, Linear programming, Inventory management, Lean manufacturing

